Production ML Pipeline

From Jupyter notebooks to real-time threat detection API

Production ML Pipeline

Transitioning your machine learning model from research to production for real-time threat detection.

High-Throughput API

Ingest live network flow data and return classifications in milliseconds

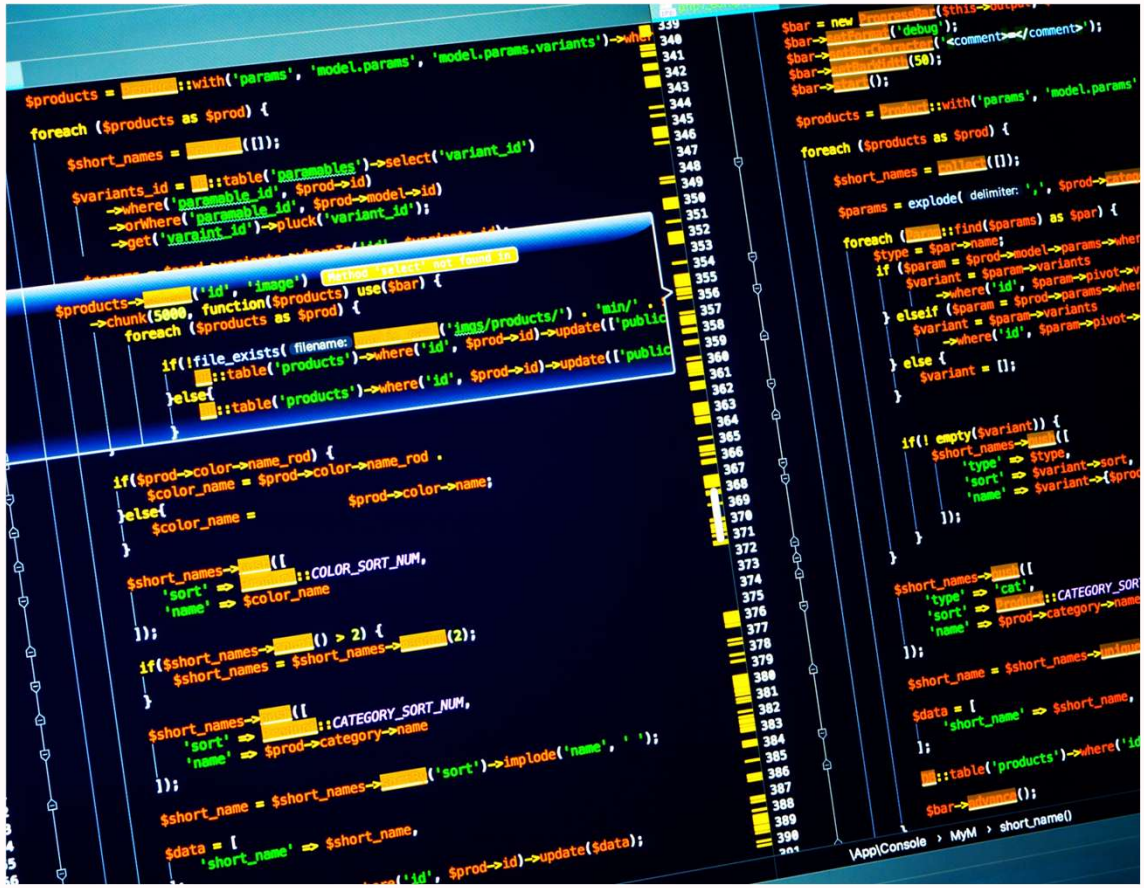
Real-Time Mitigation

Enable immediate threat response for network security

Microservices Pattern

Standalone Detection Engine scales independently based on traffic volume

Core Technology Stack



01

FastAPI Framework

High performance, asynchronous support, automatic OpenAPI/Swagger documentation

02

Model Serialization

Joblib or ONNX for faster inference

03

Containerization

Docker for consistent deployment

04

Messaging Layer

Redis or Kafka for asynchronous streams

API Endpoint Design

1

Endpoint Configuration

POST /v1/predict/threat

Accepts network flow features and returns probability scores for attack vectors

2

Request Payload

JSON with 80+ network flow features including duration, packet counts, length metrics, and idle times

3

Response Structure

Classification results with threat flag, prediction type, confidence score, risk level, and timestamp

PI Request-Response Flow (with REST and JSON

Mobile App Initiates Request

The mobile app sends an HTTP request (typically
PUT, or DELETE) to the API server.

API Server Processes Request

The API server receives the request, proc
interacts with the database.

Database Interaction

The server queries the database to fetch or upda
the required data.

Data Retrieval/Update

The database retrieves or updates the requ
data as needed.

Response Formatting

The API server formats the retrieved data into a JSO

Mobile App Parses Response

The mobile app receives and parses the JSON-f
response for display.

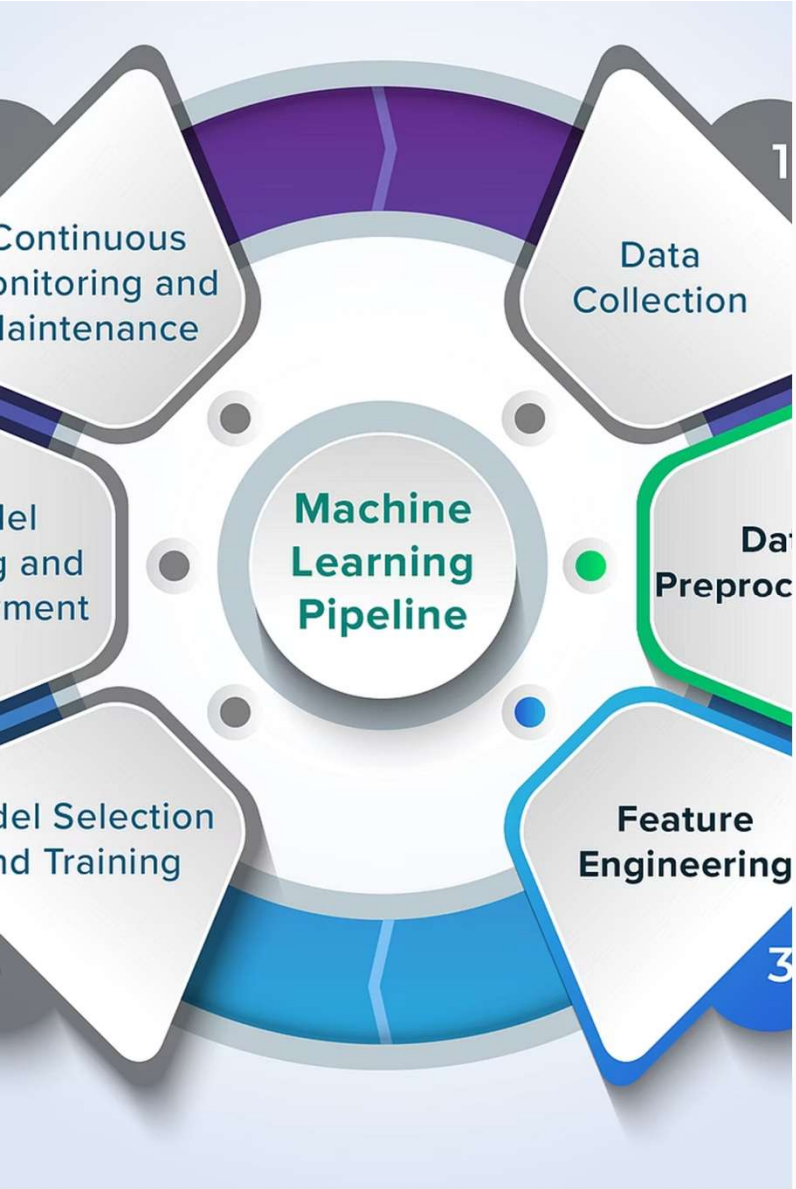
Request & Response Examples

Request Payload

```
{
  "flow_duration": 15204,
  "total_fwd_packets": 5,
  "total_bwd_packets": 3,
  "fwd_packet_length_max": 440.0,
  "idle_mean": 0.0,
  // 80+ required features
}
```

Response Payload

```
{
 "is_threat": true,
 "prediction": "DoS Hulk",
 "confidence_score": 0.987,
 "risk_level": "High",
 "timestamp": "2026-04-16T16:10:00Z"
}
```



Backend Integration Pipeline

Model Export

Pickle LightGBM model, RobustScaler, and LabelEncoder objects

Data Validation

Pydantic ensures all 84 features present and numeric

Preprocessing

Handle missing values, apply scaling, reshape for model input

Inference

Custom thresholding flags suspicious cases for review

Scalability & Performance

Optimized to handle 1.7M+ row equivalent of live traffic

Worker Processes

Uvicorn with multiple workers handles concurrent requests efficiently

Batch Inference

Process 100 flows per request to reduce overhead

Result Caching

Redis caches identical recurring flows from known safe IPs

Security & Logging

Hardened Security

Essential protections for security infrastructure

- **Authentication**

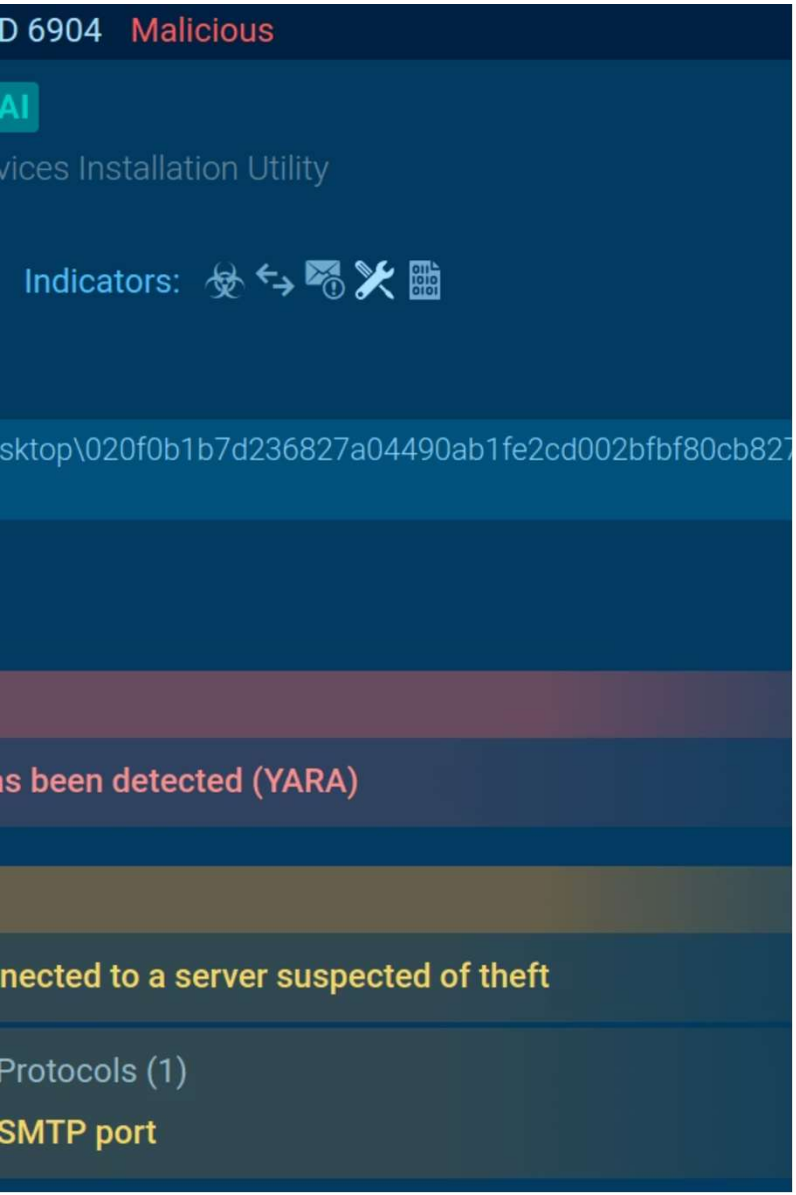
  JWT tokens or API keys ensure only authorized network monitors can query the model

- **Audit Logging**

  High-risk detections logged to PostgreSQL/Elasticsearch for forensic analysis and model retraining

- **Monitoring**

  Track inference latency; alerts trigger if detection exceeds 200ms threshold



Future: Active Response Loop

Detection Phase

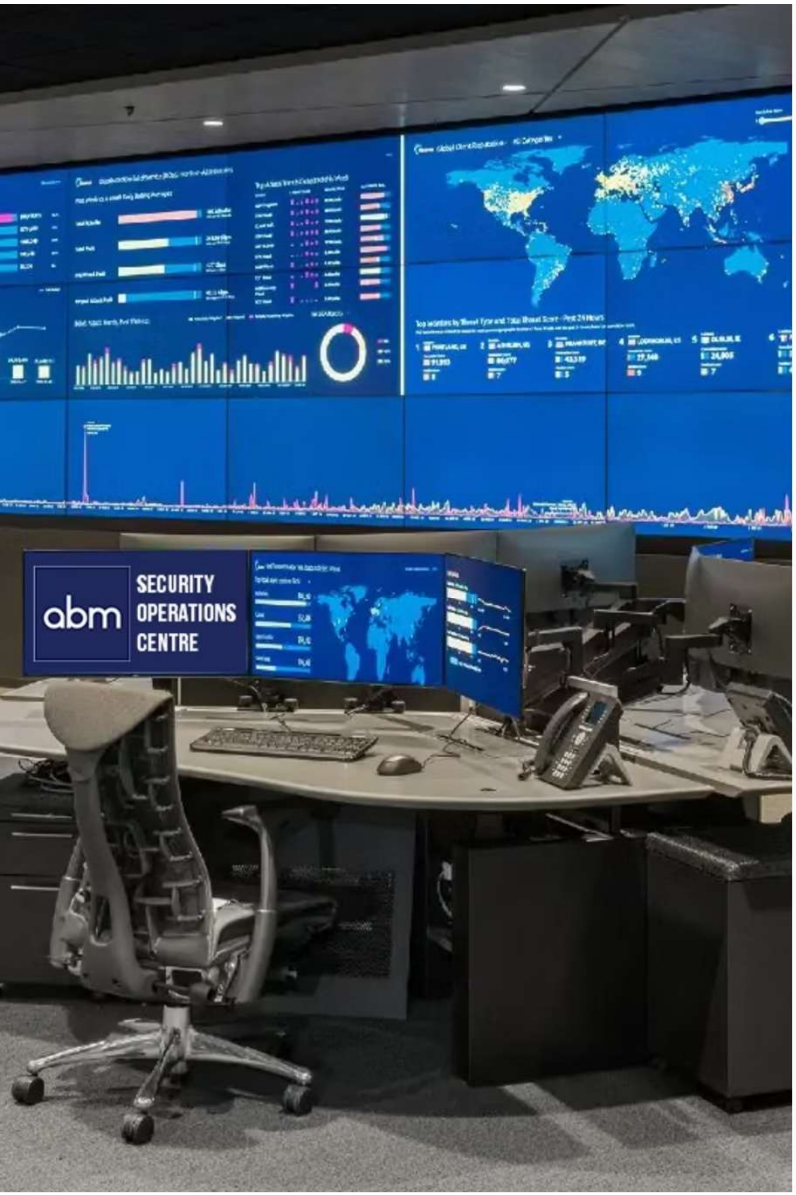
API identifies DDoS attack from specific IP address

Trigger Event

Backend sends signal to webhook endpoint

Automated Action

Script updates Cloudflare/AWS WAF rules to block IP for 60 minutes



Key Takeaways

Microservices Architecture

Independent scaling based on network traffic volume

FastAPI Framework

High performance with automatic documentation

Real-Time Processing

Millisecond classifications enable immediate threat mitigation

Security-First Design

JWT authentication, audit logging, and latency monitoring